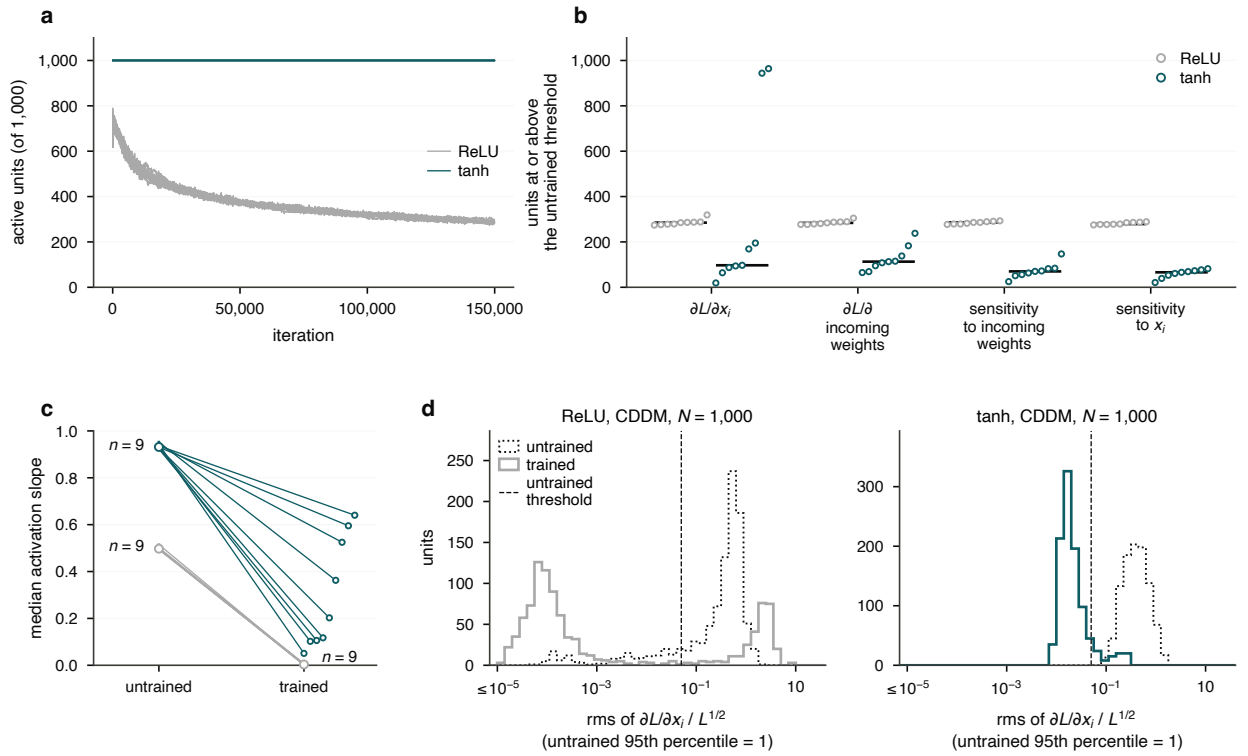

a
active units (of 1,000)
1,000
800
600
400
200
0
0
50,000
100,000
150,000
iteration
ReLU
tanh
b
units at or above the untrained threshold
1,000
800
600
400
200
0
ReLU
tanh
∂L/∂x_i
∂L/∂ incoming weights
sensitivity to incoming weights
sensitivity to x_i
c
median activation slope
1.0
0.8
0.6
0.4
0.2
0.0
n = 9
n = 9
n = 9
untrained
trained
d
ReLU, CDDM, N = 1,000
untrained
trained
untrained threshold
units
250
200
150
100
50
0
≤10^−5
10^−3
10^−1
10
rms of ∂L/∂x_i / L^1/2
(untrained 95th percentile = 1)
tanh, CDDM, N = 1,000
300
200
100
0
≤10^−5
10^−3
10^−1
10
rms of ∂L/∂x_i / L^1/2
(untrained 95th percentile = 1)